

INDIVIDUAL LOGBOOK

CSE3433 SOFTWARE ARCHITECTURE

Semester II 2025/2026

STUDENT INFORMATION

Student Name: PETER LOURDES FERNANDEZ

Matric Number: S76928

Group Number: Group 3

Project Title: Microservice – Based Disaster Relief Coordination Platform (DRCP)

Role: Full Stack & Architecture Lead

Lecturer: Dr Mohamad Nor

TEAM MEMBER AND ROLES

Name	Matric Number	Role
Muhammad Hazim Ihsan bin Adnan	S76214	Engineered the core incident Service and structured its dedicated database to securely manage emergency reports and isolate critical data
Pete Lourdes Fernandez	S76928	Envelop the central API gateway, handling request routing and seamlessly combining the independent microservice into a unified platform
Ahmad Shahiran Amsyar Bin Shafein	S74307	Built the resource Service and its corresponding database to efficient track, manage and update the inventory of relief supplier

WEEKLY PROGRESS

Phase 1 : Week 3 – Form A Group And System Overview

Weekly Progress Matric	Details & Data Logs
Group discusion	Built a group of 3 people and division of duties
Task define	Define the system scope. Identify Functional Requirements (User Management, Incident Reporting, Resource Tracking) and Non-Functional Requirements (Fault Tolerance, Scalability).
Understand the requirement and the assignment	During this phase, we analyzed the project requirements, reviewed the assignment objectives, and gained a comprehensive understanding of the overall system, including its functionality, structure, and expected outcomes
Understand the design	Try to understand the microservice architecture and what the software needed

Phase 2 – Deliverable 1 Week 5 : Project Proposal

Weekly Progress Matric	Details & Data Logs
Tasks Undertaken	Reviewed the completed proposal before submission and verified technical feasibility of proposed system functionalities.
Individual Contributions	Provided technical feedback on system requirements and implementation considerations. Assisted in validating project scope from a development perspective.
Architecture Design Draft	Create the visual diagrams. Ensure everyone understands how the services (Gateway, Incident, Alert, Logistics) will communicate via REST APIs.
Module and repository setup	Every understand their module and setup the workspace.

Week 7 – Deliverable 2: Architecture Design

Weekly Progress Matric	Details & Data Logs
Tasks Undertaken	Reviewed completed architecture diagrams and documented technical implementation considerations.
Prepared for the Delivery 2	Review and compile the final "Deliverable 2: Architecture Design" document. Ensure all diagrams, explanations, and formatting meet the rubric requirements
Challenges Faced	Understand the microservice structure and how to apply it to the service
Action Plan	Set up development environment, begin prototype implementation. And submit the report for delivery 2

Week 8 – Prototype Development

Weekly Progress Matric	Details & Data Logs
Tasks management	Divide the workload to avoid Git conflicts. Set up the GitHub repository and folder structure. Begin the Incident-service module
Start the prototype	Begin coding the prototype (incident-service). Everyone builds their basic local API endpoints (e.g., using Java Ant or Node.js) with temporary in-memory arrays (no real database yet)

Challenges Faced	Make sure the system return 201 status
Action Plan	Develop each module that already assign to every member (incident-service)

Week 9 – prototype progress

Weekly Progress Matric	Details & Data Logs
Tasks Undertaken	Continue to develop the incident-service prototype
individual Contributions	Upgrade the prototype and fixing the error, try to apply mongo db to the prototype
Challenges Faced	Try to understand the mongodb, and try to apply the rest API to the system
Action Plan	Continue development of incident service

Week 10

Weekly Progress Matric	Details & Data Logs
Tasks Undertaken	Connect the Gateway to the backend services. Use Postman to test that routing works.
Individual Contributions	Download the postman, understand it and apply to the project. Test the API using postman
Challenges Faced	The postman crash, don't understand how to use it.
Action Plan	Study the postman, fix the crush problem

Week 11 – Deliverable 3: Prototype Progress

Weekly Progress Matric	Details & Data Logs
Tasks Undertaken	Finalize the working prototype
Individual Contributions	Last upgrade the prototype. Apply the node.js and the mongodb. Help team member
Challenges Faced	The node.js give problem to the system, the mongodb still cant be apply
Action Plan	Try the best to make sure the module success,

Week 12

Weekly Progress Matric	Details & Data Logs
Tasks Undertaken	Compile the module and integrate with other service
Individual Contributions	integrate with resource-service and dispatch-service
Challenges Faced	Can not integrate with other, other service have problem
Action Plan	Test the monodb, rest API, and integrate with other

Week 13 – Deliverable 4: Project Report and Individual Logbook

Weekly Progress Matric	Details & Data Logs
Tasks Undertaken	Completed final testing and supported preparation of project documentation.
Individual Contributions	Performed final code review, validated database functionality, and ensured deployment readiness. Assisted in documenting implementation details and verifying system functionality before final submission.
Challenges Faced	Ensuring all modules functioned correctly and consistently for final evaluation.

Action Plan	Submit final project deliverables and document lessons learned throughout the development process.
--------------------	--

GENERATIVE AI USAGE RECORD

Week	Tool Used	Purpose	Prompt Used	Output Summary	How It was Used
Week 3	Gemini	Understand project overview	What is assignment is about	The overview of the assignment	Use for understand more deep about the assignment
Week 4	Gemini	Architecture Research	Explain the microservice architecture	Provided architecture explanations	Used for understand architecture basic
Week 5	Gemini	Diagram explanation	Make the diagram for this assignment.	Give simple diagram	Use for guideline for the diagram
Week 6	Gemini	Service Identification	Explain the diagram, how built the system ?	Explain the diagram more detail	Understand the diagram more deeply and finalize the diagram
Week 7	Gemini	Service Identification	Create the incident service for this assignment base on the diagram given	Give simple and working local host incident-service	Be the guideline and starting point for the prototype
Week 9	Gemini	Study the mongodb, rest API and noed.js	How to use the mongo db and how to apply the rest API	It give instruction and explain how to use it	Understand the microservice with other software
Week 11	Gemini	Prototype testing	how to know my prototype	Explain the prototype should be	Use for compare and evaluate the

			is function and ready for the final phase		system
Week 13	Gemini	Report Writing and final product integration	How integrate with other service	Explain the integration	Finishing the final product